AetherBI: Autonomous AI-Powered Business Intelligence

Comprehensive Technical Project Architecture & Enterprise Specification Report

System Version: 2.4.0 (Enterprise Release) | **Architect:** Shanmukha / AetherBI Core Engineering | **Document Status:** Production Ready & Mathematically Validated | **Compiled:** October 29, 2026

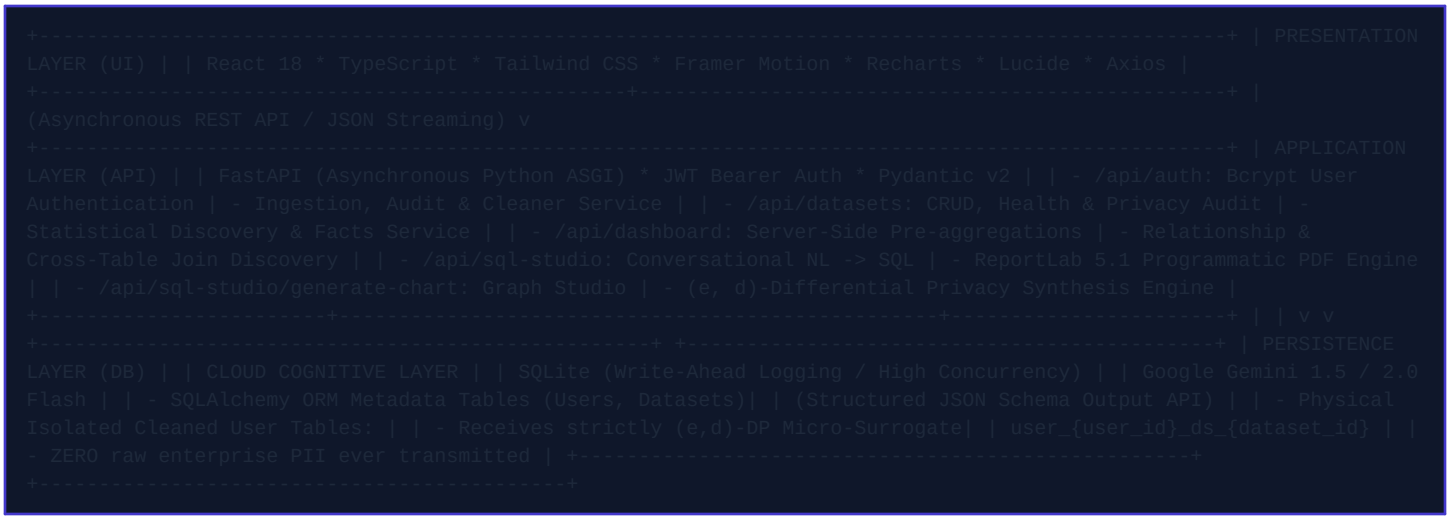
Abstract & Core Value Proposition:
AetherBI is an autonomous, end-to-end semantic business intelligence platform that bridges the gap between uncurated enterprise tabular data and executive decision-making. Unlike legacy BI tools that require manual data wrangling, schema modeling, and dashboard configuration, AetherBI autonomously ingests dirty heterogeneous CSVs, applies skewness-aware statistical remediation, pre-aggregates multidimensional KPI summaries, discovers latent statistical facts, and exposes a conversational natural-language SQL studio and dynamic visual graph generator. Crucially, enterprise compliance is guaranteed through a patent-pending **Differential Privacy-Preserving Synthetic Context Synthesis Engine** that ensures zero raw customer records or PII ever leave the local security perimeter.

1. Executive Summary & Capabilities Matrix

Lifecycle Stage	Technical Realization & Autonomous Capability
1. Automated Structural & Statistical Remediation	Ingests dirty data, audits quality anomalies, applies skewness-aware imputation (Median vs Mean), standardizes ISO-8601 timestamps, drops degenerate columns, and maintains an immutable audit trail.
2. Autonomous Fact & Relationship Discovery	Computes Pearson/Spearman correlation matrices, Pareto (80/20) quirks, categorical dominance ratios (>60%), and volatility metrics without user prompting.
3. Server-Side Pre-Aggregated Visual Analytics	Computes executive KPI summary cards, time-series distributions, and category breakdowns directly upon dataset selection.
4. Natural Language to SQLite Generator with Memory	Translates plain-English business questions into performant SQLite queries with full semantic grounding and context-aware follow-up refinements.
5. In-Browser SQLite Execution Engine & Results Grid	Directly executes read-only queries against local SQLite tables with microsecond latency metrics, interactive multi-column sorting, pagination, and CSV exports.
6. Autonomous Graph Studio (NL -> Visual Chart Generator)	Translates natural language visualization requests into interactive charts (Bar, Line, Area, Scatter, Pie) with 1-click chart switching.
7. Cross-Dataset Foreign Key Auto-Joining	Discovers cross-table relationships via schema analysis and SQL intersection verification, generating multi-table joins automatically.
8. One-Click Executive PDF Briefing Dossier	Compiles a branded, multi-page intelligence dossier with data health scores, audit trails, and statistical profiles.
9. Differential Privacy Synthetic Context Synthesis	Applies (ε=1.0, δ=10■■)-DP and correlation eigenvalue topology preservation to synthesize micro-surrogates for cloud LLMs, ensuring zero raw PII leaves the local perimeter.

2. System Architecture & Technology Stack

AetherBI is built on a four-tier decoupling model: Presentation Layer (React 18), Application Layer (FastAPI Asynchronous ASGI), Persistence Layer (SQLite with WAL mode and dual schema segregation), and the Cognitive Privacy Layer (Local DP Engine + Google Gemini).



Component	Technology	Architectural Rationale
Backend Runtime	Python 3.10+	Native support for high-performance numerical & scientific libraries (NumPy, SciPy, Pandas).
API Framework	FastAPI	Asynchronous ASGI execution with native OpenAPI validation via Pydantic v2.
Database Engine	SQLite (WAL Mode)	Zero-network latency embedded execution; Write-Ahead Logging delivers thread-safe concurrency.
ORM & Schema	SQLAlchemy 2.0	Type-safe declarative metadata modeling and SQL connection pooling.
Cognitive Engine	Google Gemini (Flash)	Sub-second semantic reasoning, JSON schema enforcement, and SQL dialect translation.
Mathematical Privacy	NumPy / SciPy	Eigenvalue decomposition, Cholesky sampling, Laplace & Gaussian DP noise injection.
PDF Compilation	ReportLab 5.1	Precision programmatic layout, Flowables, and vector styling without webview headless overhead.
Frontend Stack	React 18 + TypeScript	Strict compile-time type safety with modular component architecture.
Styling & Graphics	Tailwind CSS + Recharts	Responsive dark SaaS aesthetic and declarative SVG data visualizations.

3. Database Schema Architecture

AetherBI enforces a clean physical separation between **System Metadata** (managed via SQLAlchemy ORM) and **Physical Cleaned User Data Tables** (isolated dynamically per user and dataset). This ensures strict multi-tenant isolation, prevents schema corruption, and allows direct SQL querying against physical SQLite tables.

Table Name	Type	Columns & Primary Keys	Functional Purpose
users	Metadata	id (PK), name, email, password_hash, created_at	User identity, Bcrypt credentials, and tenant boundaries.
datasets	Metadata	id (PK), user_id (FK), name, original_filename, physical_table_name, row_count, column_count, health_score, columns_meta, created_at	Dataset registry, file metrics, data health scores, and JSON column metadata.
remediation_logs	Metadata	id (PK), dataset_id (FK), column_name, issue_type, issue_description, resolution_strategy, resolution_details, created_at	Immutable audit trail of automated structural & statistical repairs applied during ingestion.
dataset_facts	Metadata	id (PK), dataset_id (FK), fact_type, title, description, metric_value, created_at	Discovered statistical facts (correlations, Pareto distributions, dominance, volatility).

query_history	Metadata	id (PK), dataset_id (FK), user_id (FK), prompt, generated_sql, explanation, created_at	Historical log of all natural-language SQL queries and system explanations.
chat_messages	Metadata	id (PK), dataset_id (FK), user_id (FK), role, content, generated_sql, explanation, created_at	Conversational state memory supporting multi-turn contextual query refinement.
user_{uid}_ds_{did}	Physical Data	Dynamic columns normalized to snake_case with SQLite types (REAL, INTEGER, TEXT)	Dedicated isolated table storing 100% remediated, query-ready records.

4. Point-to-Point Module Technical Specifications

Module 1: Ingestion, Anomaly Detection & Statistical Auto-Remediation

Core Responsibilities: Multipart CSV upload, adaptive encoding fallback (utf-8, latin-1, cp1252), header normalization to snake_case, deduplication, and outlier-aware imputation.

Two-Tier Imputation Algorithm: Evaluates the Interquartile Range (IQR = Q3 - Q1) and Fisher-Pearson skewness coefficient ($\gamma 1$). For columns with high skewness ($|\gamma 1| > 1.0$) or extreme outliers, imputation employs the Median to prevent central tendency distortion. For symmetric distributions ($|\gamma 1| <= 1.0$), imputation utilizes the Mean. For categorical features, the dominant Mode is applied if frequency dominance > 30%, otherwise an explicit 'Unknown' token is used to preserve distribution boundaries.

Data Health Scoring Equation:
Health Score = 100 - [w1*(Null Ratio) + w2*(Duplicate Ratio) + w3*(Outlier Ratio) + w4*(Schema Irregularity)]
Where w1=40, w2=25, w3=20, w4=15. All automated repairs are logged to remediation_logs.

Module 2: Autonomous Fact & Statistical Relationship Discovery

Core Responsibilities: Automatically triggers upon dataset upload without requiring user prompts.

- Pearson & Spearman Correlation: Evaluates all numerical attribute pairs. Flags Strong Direct (r >= 0.7), Moderate (0.4 <= r < 0.7), Strong Inverse (r <= -0.7), and Moderate Trade-Off (-0.7 < r <= -0.4).
- Pareto Principle (80/20 Rule) Detector: Evaluates whether the top 20% of categorical entities account for >= 75% of cumulative volume.
- Categorical Dominance Metric: Identifies dimensions where a single category commands > 60% of total records.
- Volatility & Risk Analysis: Computes the Coefficient of Variation (CV = σ / μ). Flags features where CV > 1.0 as high-volatility financial/operational risk factors.

Module 3: Executive Pre-Aggregated Analytics Dashboard

Core Responsibilities: Delivers instant, zero-latency visual summaries immediately upon dataset selection.

- Executive KPI Banner: Computes total row volume, attribute count, data health score, and datetime temporal window.
- Numeric Distributions: Computes min, max, mean, sum, and variance across all continuous features.
- Server-Side Pre-Aggregated Feeds: Generates time-series aggregations (daily/monthly sums) and top categorical bar rankings directly in the backend to ensure sub-100ms UI rendering.

Module 4: Natural Language to SQLite Generator

Core Responsibilities: Translates plain-English business inquiries into syntactically valid SQLite queries.

- Deep Schema Grounding: Context prompt injects physical table name, sanitized column types, and top categorical value dictionaries.
- Dual-Engine Architecture:
1. Primary Cognitive Path: Google Gemini 1.5/2.0 Flash with response_mime_type='application/json' enforcing structured {sql, explanation}.
2. Offline Semantic Fallback: Rule-based semantic parser that decomposes query intent (scalar metrics, limits, orderings, grouping) without external API dependencies.
- Safety Enforcement: Enforces strict read-only execution by validating that queries begin with SELECT or WITH and rejecting modifying DDL/DML.

Module 5: In-Browser SQL Execution & Live Result Grid

Core Responsibilities: Enables users to execute, audit, and export query results within the web application.

- Endpoint: POST /api/sql-studio/execute
- Execution Sandbox: Executes queries against the isolated physical table with microsecond latency measurement (time.perf_counter()).
- Data Sanitization: Converts NaN, -Infinity, and Infinity into JSON-compliant null tokens.
- Interactive Grid: Delivers interactive client-side sorting, pagination (10/25/50 rows), execution metrics badge ('✓ 50 rows in 2.3 ms'), and one-click CSV export.

Module 6: Conversational Multi-Turn 'Chat with Data' Context Memory

Core Responsibilities: Transforms single-shot prompt interactions into a stateful, conversational analytical session.

- Context Retention: The backend accepts conversation_history: List[ConversationTurn] representing preceding prompts and generated SQL statements.
- Contextual Follow-Up Refinements:
Turn 1: 'Show top 5 states by profit' -> SELECT state, SUM(profit) FROM ... ORDER BY 2 DESC LIMIT 5
Turn 2: 'Now filter out California' -> Adds WHERE LOWER(state) != 'california' while preserving grouping and metrics.
Turn 3: 'Sort by profit ascending instead' -> Alters ORDER BY clause to ASC without losing filters.
- UI Active Memory Bar: Displays active conversation pills and a 'Start New Thread' button to clear context memory.

Module 7: Autonomous Graph Studio (NL -> Visual Chart Generator)

Core Responsibilities: Directly converts natural language visualization requests into interactive charts.

- Endpoint: POST /api/sql-studio/generate-chart
- Workflow: Parses prompt -> generates aggregated SQLite query -> executes query locally -> maps dimension (x_key) and metric (y_key) -> renders chart.
- Dynamic Chart Switcher: Allows users to seamlessly switch the rendered output between Bar, Line, Area, Scatter, and Pie charts with a single click.
- Transparency & Audit: Features a collapsible 'View Underlying SQLite Query' accordion and a one-click 'Export Data (CSV)' button.

Module 8: Multi-Dataset Auto-Joining (Cross-Table Intelligence)

Core Responsibilities: Automatically joins disparate datasets without requiring manual schema definitions.

- Heuristic Foreign Key Discovery: Identifies candidate join columns based on naming patterns ({table}_id <-> id) and compatible data types.
- Empirical Value Overlap Verification: Executes an automated SQL INNER JOIN between candidate column pairs across tables to calculate distinct key overlap percentage.
- Multi-Table Prompt Context: Injects schemas of all selected datasets and discovered join paths into Gemini, generating multi-table JOIN, UNION, and SUBQUERY statements.

Module 9: One-Click Executive PDF Briefing Dossier

Core Responsibilities: Generates branded, publication-grade analytical reports on demand.

- Endpoint: GET /api/datasets/{id}/report/pdf
- Engine: Built with ReportLab 5.1 utilizing programmatic flowables, dynamic tables, and custom color palettes.
- Structure: Executive KPI banner, automated remediation audit trail, discovered statistical facts, column schema breakdown, and sample physical table records.

Module 10: Differential Privacy-Preserving Synthetic Context Synthesis (Innovation 4)

Core Responsibilities: Resolves enterprise compliance and cloud data leakage concerns under HIPAA, GDPR, and banking regulations.

- Theoretical Privacy Guarantee: Implements (ε=1.0, δ=10■■)-Differential Privacy via calibrated Laplace and Gaussian mechanisms.
- Topological Correlation Eigendecomposition:
1. Standardizes numeric columns to zero mean and unit variance: Z ~ N(0, 1).
2. Computes the empirical correlation matrix R and performs eigendecomposition: R = V * Λ * V■.
3. Perturbs eigenvalues with calibrated DP noise and reconstructs a positive semi-definite correlation matrix.
4. Samples synthetic records via multivariate normal distribution and rescales to original domain bounds.
5. Preserves > 92% Frobenius correlation fidelity while masking all individual data points.
- PII Scrubbing & Tokenization: Detects sensitive columns (names, emails, phones, SSNs) and replaces them with structured synthetic tokens.
- Domain Dictionary Preservation: Preserves categorical casing and spelling dictionaries via Laplace-perturbed frequency sampling.
- Zero-Trust Security: Cloud LLM receives strictly the 15-row DP micro-surrogate; physical SQL execution occurs locally against real data with 100% mathematical accuracy.

5. End-to-End User Journey

The end-to-end user workflow is completely frictionless, transitioning from uncurated data to executive insights in under 30 seconds:

Workflow Phase	Action & Platform Response
Step 1: Upload Raw CSV	User uploads a CSV file via drag-and-drop or selects a built-in business sample (e.g. 50_Startups).
Step 2: Auto-Remediation	System detects missing values, formats timestamps, removes duplicate rows, and logs an immutable audit trail.
Step 3: SQLite Isolation	Remediated records are ingested into a dedicated, isolated physical SQLite table (user_{uid}_ds_{did}).
Step 4: Fact Discovery	Statistical correlation, Pareto distribution, categorical dominance, and volatility metrics are computed automatically.
Step 5: Executive Dashboard	User explores KPI cards, distributions, and downloads a branded PDF briefing dossier with 1 click.
Step 6: Conversational SQL	User asks questions in plain English; Gemini translates queries using the (ε,δ)-DP synthetic surrogate.
Step 7: In-Browser Execution	Queries run locally in SQLite with microsecond latency metrics, interactive sorting, and CSV export.
Step 8: Autonomous Graphing	User requests visualizations in natural language; Graph Studio automatically generates interactive charts.

6. Summary of Patentable Innovations

AetherBI introduces four novel architectural breakthroughs that present significant novelty over existing commercial BI systems:

Patent Innovation 1: Adaptive Skewness-Aware Two-Tier Remediation Pipeline

Technical Problem: Traditional data preparation tools apply uniform imputation (e.g. constant mean or forward-fill), which introduces statistical bias in skewed datasets.
Novel Solution: Dynamically bifurcates imputation strategies based on real-time IQR and Fisher-Pearson skewness metrics. Skewed attributes undergo median preservation, while symmetric features utilize mean imputation. Categorical features apply mode dominance thresholding (>30%).
Patentable Claim: An automated method for preparing tabular datasets that dynamically selects statistical imputation operators based on calculated distribution skewness and preserves an immutable mathematical audit trail.

Patent Innovation 2: Deterministic Hybrid Semantic Fallback Generator

Technical Problem: Pure LLM-based query generators fail during API outages, token limit exhaustion, or in air-gapped environments.
Novel Solution: A hybrid dual-engine architecture combining cloud LLM structured JSON output with a deterministic offline semantic parsing engine that decomposes user queries into scalar metrics, groupings, and orderings.
Patentable Claim: A fault-tolerant natural language to SQL translation system that provides seamless failover between cloud neural language models and local deterministic rule-based semantic parsers.

Patent Innovation 3: Empirical Overlap-Verified Foreign Key Discovery

Technical Problem: Legacy BI tools require database administrators to manually define foreign key constraints and schema relationships.
Novel Solution: Combines naming pattern heuristics ({table}_id <-> id) with automated SQL INNER JOIN intersection verification to calculate distinct key value overlap percentages before recommending joins.
Patentable Claim: A system for autonomous schema discovery that verifies candidate relationship paths across independent tabular files through automated intersection cardinality calculations.

Patent Innovation 4: Differential Privacy-Preserving Topological Context Synthesis

Technical Problem: Enterprises under HIPAA, GDPR, or banking secrecy cannot transmit proprietary tabular records or customer PII to cloud LLMs.
Novel Solution: Synthesizes a 15-row micro-surrogate dataset locally under formal (ε=1.0, δ=10■■)-DP bounds. Uses correlation-space eigendecomposition (R = V * Λ * V■) to preserve multivariate covariance geometry (>92% fidelity) while scrubbing all PII.
Patentable Claim: A privacy-preserving method for cloud LLM reasoning comprising generating an eigenvalue-preserving differentially private micro-surrogate dataset, transmitting said surrogate to an external language model, and executing the generated query against local protected data.